

Obesity as a catalyst for neurodegeneration

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Bandy Chen¹✉, Amanda Rodríguez-Díaz², Marc Schneeberger²✉ & Eric Topol³✉

The parallel rise in obesity and neurological disorders suggests a potential mechanistic link between these two conditions. Midlife obesity is a well-established risk factor for cognitive decline and neurological disorders; however, the underlying mechanisms and cause-and-effect relationship remain poorly defined, in part owing to limited spatiotemporal resolution in existing studies and the absence of a unifying framework. Recent studies have begun to uncover pathways linking obesity to neurodegenerative vulnerability. First, we discuss how obesity induces brain-wide reprogramming across neural, metabolic and vascular systems including neurovascular coupling, blood–brain barrier integrity, cerebrospinal fluid dynamics and myelination. By systematically examining the disparate mechanisms for each condition, a more cohesive picture emerges, revealing progressive neurodegeneration driven by chronic metabolic overload. Lastly, we propose a framework that integrates these different mechanisms with both spatial and temporal specificity.

“Every solution to a problem raises new unsolved problems.” – Karl Popper

The industrialization of the global food system began with a well-intentioned goal: to combat hunger and food scarcity on a global scale. Over time, this effort led to the mass production of ultra-processed, calorie-dense and highly palatable foods. These products contributed to an increased food security and longer lifespans but also introduced new challenges. The modern nutritional landscape now fuels chronic diseases that erode long-term metabolic, cardiovascular and cognitive health¹. For the first time in human history, overnutrition, rather than scarcity, has become a leading driver of disease. The rapid surge in obesity places immense pressure on public health systems to develop prevention and treatment strategies. Current evidence indicates that increased calorie intake, rather than reduced physical activity, is considered the root cause of the global obesity epidemic².

Alongside obesity, the prevalence of type 2 diabetes mellitus (T2DM), metabolic dysfunction and neurodegenerative disorders has grown at an alarming rate. This parallel rise has sparked renewed interest in how chronic exposure to hyperpalatable, metabolically disruptive diets may rewire the brain's homeostatic and reward circuits, reinforcing cycles of overeating and physiological dysregulation³. The epidemiological trajectory suggests that metabolic overload and

neurological vulnerability are not isolated phenomena but may reflect shared biological mechanisms⁴.

A growing body of evidence supports this view. Chronic metabolic overload alters central circuits that maintain energy balance, disrupts neurovascular and inflammatory homeostasis and accelerates brain ageing⁵. Obesity is consistently associated with reduced cognitive performance and brain volume, and imaging signatures of accelerated brain ageing^{6,7}. Unsurprisingly, these accumulating alterations heighten the risk for neurological disorders⁸. Importantly, obesity is not a uniform state. Visceral adiposity, defined as the excess intra-abdominal fat surrounding internal organs, is the strongest predictor of metabolic dysfunction, more so than total fat mass, and is linked to greater structural and functional disruption within the brain^{9,10}. The mechanisms discussed in this Perspective, including inflammatory, vascular-barrier and fluid-structural pathways, primarily reflect the consequences of visceral obesity.

Yet a shared mechanism does not automatically imply causality or a unified disease model. To understand how chronic metabolic overload contributes to progressive neural decline, both reductionist and systems-level approaches are needed. This Perspective adopts that dual framework: dissecting how obesity perturbs specific cellular, vascular and inflammatory processes (reductionist view), while

¹Department of Medicine, UC San Diego School of Medicine, La Jolla, CA, USA. ²Laboratory of Neurovascular Control of Homeostasis, Department of Cellular and Molecular Physiology, Yale University School of Medicine, New Haven, CT, USA. ³Scripps Research Translational Institute, La Jolla, CA, USA.

✉e-mail: bac008@health.ucsd.edu; marc.schneebergerpane@yale.edu; etopol@scripps.edu

assembling these alterations into an integrated model of how metabolic stress accelerates neurodegeneration (systems-level view). Throughout this Perspective, we clearly differentiate between pre-clinical and clinical evidence and explicitly note where human data remain limited.

As we uncover how metabolism sculpts the brain at the molecular, cellular and systems levels, a holistic picture emerges: metabolic health is not only about the body; it is fundamentally about the brain. Understanding this relationship is essential for designing the next generation of metabolic therapies, in which protecting the brain is as central as controlling weight or glucose.

Obesity-driven vulnerability to cognitive decline and neurodegeneration

Although obesity is associated with measurable cognitive impairments, the extent to which these changes reflect actual neurodegeneration processes remains unresolved. Obesity and neurodegeneration do not constitute a strict clinical continuum. Most individuals with obesity will never develop a neurodegenerative disorder, and many people with Alzheimer's disease or related conditions have no history of obesity. Nevertheless, epidemiological evidence consistently links metabolic dysfunction to an increased risk of accelerated cognitive ageing¹¹, suggesting that obesity can act as a vulnerability factor in certain individuals.

Large-scale neuroimaging cohorts have reported overlapping patterns of brain atrophy in obesity and Alzheimer's disease, particularly in temporo-parietal and hippocampal regions¹². Dietary patterns provide further insight: higher consumption of ultra-processed foods, rich in refined carbohydrates, fats and additives, is associated with accelerated cognitive decline¹³, whereas diets enriched in whole foods, such as vegetables, fruit and whole grains, correlate with greater cortical and hippocampal volumes¹⁴. This relationship appears especially salient in adolescence, a period of heightened neurodevelopmental plasticity, when childhood obesity has become a global health concern¹⁵. Cognitive deficits, such as sluggish cognitive tempo, are more prevalent in children with obesity than in their healthy peers¹⁶, raising concern that early-life metabolic stress may reshape long-term neurocognitive trajectories. Maternal obesity compounds this vulnerability, with increased rates of neuropsychiatric and neurodevelopment disorders, documented across offspring cohorts^{17,18}. Together, these data underscore the urgency of public health strategies aimed at mitigating childhood and maternal obesity to reduce downstream neurological burden¹⁹.

While epidemiological associations highlight the scope of the problem, experimental models have begun to reveal the biological pathways that may causally link obesity to neural vulnerability. Pre-clinical studies provide more mechanistic consistency than human data, yet their translational relevance must be interpreted cautiously. In mouse models, obesity induces glial senescence within neurogenic niches, and clearance of these senescent cells restores neurogenesis and ameliorates anxiety-like behaviour²⁰. Obesity also heightens neuroinflammation, promotes lipid accumulation in glial and endothelial populations, and compromises blood–brain barrier (BBB) integrity, collectively sensitizing the brain to injury and degeneration²¹. Notably, metabolic overload exacerbates cognitive decline across diverse models of Alzheimer's disease and Parkinson's disease^{22–24}, suggesting that obesity can accelerate pathogenic processes even when it is not the primary driver of disease.

Therefore, emerging evidence supports a view in which obesity exerts widespread systemic effects, including metabolic, inflammatory, vascular and hormonal effects, that converge on the brain to lower its threshold for dysfunction^{4,25,26}. This Perspective synthesizes these insights to examine how chronic metabolic overload reshapes cerebral architecture and function, increasing susceptibility to cognitive decline and neurological disease.

The domino effect of neurodegeneration

Understanding how neurocircuit dysfunction unfolds in the context of obesity requires moving beyond the traditional neuron-centric model, where neurons are seen as the sole drivers of brain function. Instead, a more comprehensive framework is emerging in which glial, vascular and immune cells have indispensable and interconnected roles in maintaining neural stability²⁷. Within this expanded view, the brain is recognized as a dynamic neuro–glial–vascular ecosystem whose homeostasis depends on coordinated communication across multiple cell types. Chronic metabolic overload disrupts this balance, provoking time-dependent and region-specific shifts in cellular states. By examining how signals propagate across these diverse populations, we can begin to identify network-level vulnerabilities that predispose specific brain circuits to neurodegeneration.

Recent advances in neuroscience, including single-cell and spatial omics, genetically defined cell-type-specific targeting and intravital imaging, have enabled high-resolution mapping of cell–cell interactions within the brain^{28,29}. These tools make it possible to move beyond isolated pathways and towards an integrated understanding of how obesity perturbs the broader architecture of brain function. The following section introduces an emerging paradigm that shifts from a neuron-centric view and integrates the entirety of the neuro–glial–vascular unit (NVU), capturing the microenvironmental complexity and heterogeneity that shape circuit resilience.

A well-characterized example of NVU-mediated communication is neurovascular coupling (NVC) or functional hyperaemia, in which active neurons signal to nearby blood vessels to increase local cerebral blood flow (CBF)³⁰. Yet neuron–vascular interactions extend far beyond NVC. Neuronal activity also drives vascular remodelling, modulates glymphatic clearance of metabolic waste and influences oligodendrocyte maturation and adaptive myelination^{31–34}. These processes are deeply interdependent, forming an intricately coordinated network in which perturbations at any node can reverberate through the system. Disruption of this finely tuned coordination has profound implications for both local circuit function and global network integrity.

At the systems level, interventions such as brain-wide gamma stimulation illustrate the integrative nature of NVU function: these oscillatory manipulations simultaneously promote vascular remodelling, glial plasticity, waste clearance and myelin adaptation^{35–38}. Such findings underscore that brain health is inherently multicellular, sustained not by neurons alone but by the emergent properties of a coordinated, brain-wide cellular ensemble. When challenged by chronic metabolic overload, this intricate system becomes increasingly fragile, initiating a cascade of maladaptive responses that propagate through neurons, glia and vasculature in a progressive, circuit-specific manner.

In the following sections, we adopt a reductionist approach to dissect how obesity alters the major components of brain homeostasis. We then integrate these insights into a cohesive systems-level framework, proposing a convergent-cascade model in which dysfunction of the NVU represents a central pathway linking chronic overnutrition to progressive neurodegeneration.

Contrasting frameworks of neurodegenerative vulnerability in obesity: cascade versus convergence

Although obesity is known to alter both brain structure and function, integrating these diverse effects into a unified mechanistic model remains challenging. The current understanding is that the brain, particularly the hypothalamus, defends a biologically stable level of adiposity through long-term homeostatic mechanisms that match energy intake to expenditure³⁹. In obesity, this defended level shifts upwards rather than collapsing entirely, and this resetting is accompanied by hypothalamic inflammation and gliosis^{40,41}. Because these exact glial and immune mechanisms are implicated in neurodegenerative diseases⁴², such persistent adaptations represent only one of several

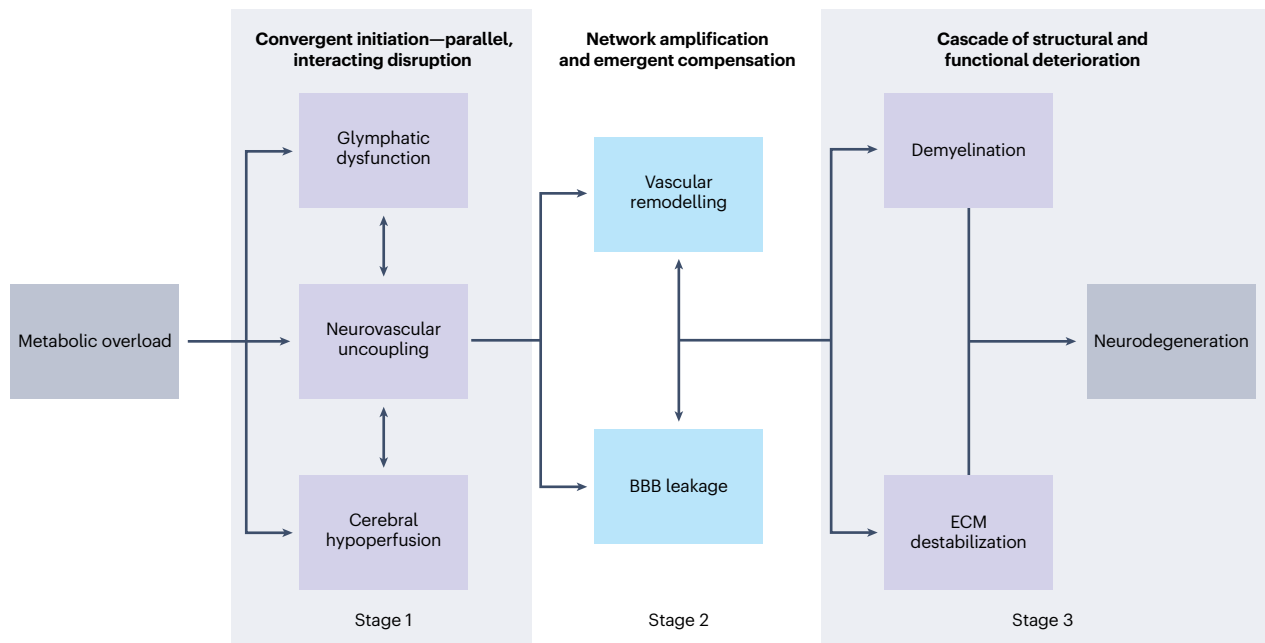


Fig. 1 | Integrated framework for obesity-induced neurodegeneration. The integrated framework aims to combine both parallel and sequential pathophysiological processes in the metabolically compromised brain. Instead of a linear cascade or isolated convergence, this model represents a dynamic and interconnected system. The first stage represents a converging set of parallel insults that independently respond to overnutrition yet are functionally interdependent. These triggers lead to adaptive responses seen in stage 2 such as vascular remodelling and increased BBB permeability. While these early changes in stage 2 can help maintain perfusion or allow more immune

surveillance and metabolic exchange, their persistence drives a maladaptive shift characterized by chronic inflammation, neurovascular uncoupling, ECM destabilization and demyelination. This reflects the cascade hypothesis, where convergent disturbances trigger sequential downstream pathologies. Stage 3 is the downstream degenerative phase where compensatory and converging processes result in potentially irreversible structural and functional damage. The cumulative effects of these stages act as a nexus between obesity and increased risk of neurodegeneration. Created in BioRender. Laboratory, S. <https://biorender.com/ya76dqy> (2026).

potential routes by which metabolic stress may reshape brain function and increase susceptibility to neural decay.

Two prevailing conceptual models have been used to interpret neurodegenerative vulnerability in obesity. The ‘cascade hypothesis’ posits that a single initiating insult, such as inflammation, triggers a sequential chain of downstream events (‘domino effect’) culminating in cognitive decline. The ‘convergence hypothesis’ suggests instead that multiple pathological processes (vascular, immune, metabolic, structural) act in parallel on shared neural substrates, ultimately converging on degenerative outcome. Canonical examples are the amyloid cascade hypothesis and the two-hit model^{43,44}. Both frameworks capture elements of disease progression yet oversimplify the complexity and interdependence of chronic metabolic injury.

In obesity, early hypothalamic inflammation has been interpreted as a potential cascade trigger. Rodents exposed to a high-fat diet (HFD) for as little as 3 days exhibit reactive gliosis and neuronal injury in the mediobasal hypothalamus. This phenomenon is also observed in humans with obesity^{45,46}. Early immune activation perturbs NVC and CBF, processes tightly regulated by glial and endothelial cells^{47,48}. Disruption of NVU communication may, therefore, initiate a self-amplifying sequence of maladaptive responses, including extracellular matrix (ECM) remodelling, myelin disruption and deficits in cognition.

The convergence hypothesis offers an alternative view, arguing that overnutrition simultaneously disrupts multiple systems, each of which contributes independently to neurodegeneration. Yet given the extensive cross-talk among NVU components, few processes operate in true isolation. Perturbations in vascular signalling rapidly influence glial states; inflammatory activation alters barrier function; metabolic stress reshapes neuronal firing. Thus, what appears as parallel pathology at the tissue level may, at the cellular level, represent a deeply interconnected system undergoing progressive decoupling.

We propose a unified ‘convergent-cascade’ framework that reconciles both models. Early metabolic overload produces a set of parallel disruptions (altered NVC, reduced CBF, impaired fluid clearance) that arise concurrently and may initially be compensatory. Over time, accumulated dysfunction reaches a tipping point (‘convergence’), after which vascular stress, glial activation and barrier breakdown begin to amplify each other in a non-linear manner⁴⁹. This escalation produces a downstream ‘cascade’ of maladaptive outcomes, including ECM destabilization, myelin loss and network-level impairment (Fig. 1). This model better reflects the multifactorial nature of obesity and aligns with analogous debates in metabolic biology, such as carbohydrate–insulin versus energy–balance models, where overlapping mechanisms reinforce one another over time.

In the next sections, we dissect how metabolic overload perturbs individual components of brain homeostasis and then integrate these findings into a cohesive systems-level framework of obesity-related neurodegeneration.

Disrupted neurovascular communication in obesity

NVC has a central role in maintaining cerebral homeostasis. It matches local neuronal demand with appropriate nutrient delivery and supports waste-clearance mechanisms⁵⁰. Disruption of this tightly regulated mechanism has been observed across neurological disorders and is frequently accompanied by alterations in CBF⁵¹. Chronic overnutrition predictably perturbs both NVC and CBF, probably reflecting an initial adaptive response to sustained metabolic excess that becomes maladaptive over time^{52,53}.

Human studies consistently demonstrate that higher body mass index (BMI) is associated with global reductions in CBF and diminished cerebrovascular reactivity, particularly in regions implicated in Alzheimer’s disease pathology, suggesting a mechanistic pathway

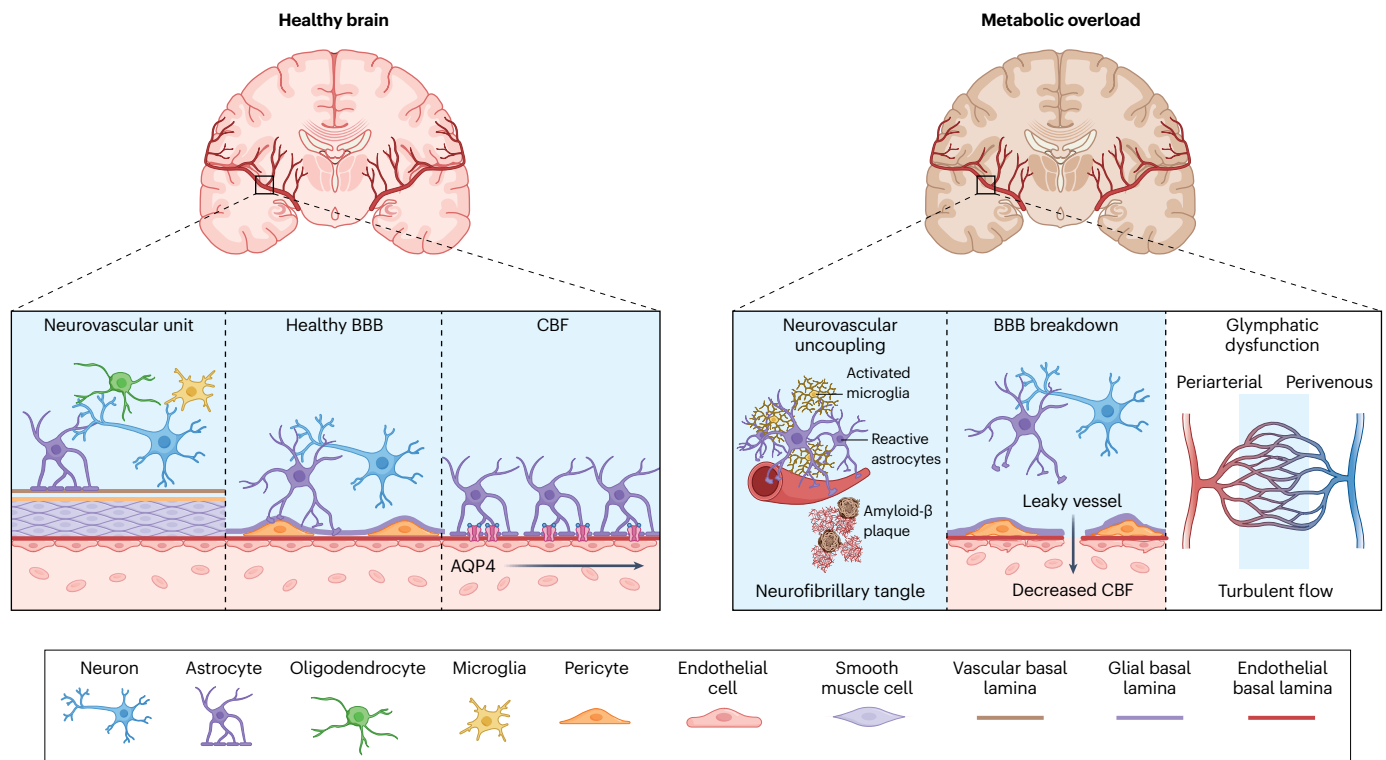


Fig. 2 | Convergent initiation—parallel, interacting disruption. Metabolic overload blunts NVC, impairs CBF and disrupts glymphatic flow via shared endothelial, astroglial and haemodynamic mechanisms. Although the temporal order of these responses is unclear, NVC is intimately linked to both CBF regulation and CSF flow. Because NVC governs activity-dependent arteriole

dilation and vasodynamics that pump CSF into perivascular spaces, its impairment can simultaneously blunt CBF and glymphatic transport. Consistent with this coupling, functional hyperaemia evoked by whisker stimulation increases periarterial CSF influx. This is driven by neural activity-dependent arterial diameter changes, rather than a direct neural modulation on CSF flow.

linking metabolic overload to cognitive decline^{54–56}. Dietary interventions reinforce this relationship: adherence to a Mediterranean diet increases whole-brain perfusion and shifts amyloid biomarkers towards a lower-risk profile measured by increased $A\beta_{42}/A\beta_{20}$ ratio⁵⁷, whereas Western-style diets have the opposite effect^{54,58,59}. Although such biomarker changes should not be interpreted as true modification of Alzheimer's disease risk, which develops over decades, they demonstrate how diet can acutely shape cerebrovascular physiology.

Physical activity exerts protective effects on cerebrovascular function in individuals with obesity⁵⁸. Notably, neurovascular alterations also emerge during adolescence, where obesity is associated with reduced CBF in the cerebellum, precentral gyrus and supplementary motor area⁶⁰. Similar reductions are seen in individuals with T2DM, particularly within the hippocampus, occipital and parietal cortices, and amygdala^{61,62}. These data point to a global shift in brain haemodynamics in response to metabolic shifts.

Converging evidence from animal models strengthens this interpretation. In HFD-fed mice, overnutrition impairs sensory-evoked CBF in the somatosensory cortex and decreases resting and odour-evoked CBF in the olfactory bulb^{63,64}. Given the olfactory bulb's extensive projections to emotion- and memory-related circuits⁶⁵, olfactory loss is a known early marker of neurological decline, and vascular dysfunction in this region may represent an early sentinel of broader neural vulnerability⁶⁶. The hippocampus is similarly affected: a HFD reduces hippocampal CBF in mice, which is associated with both visceral adiposity and cognitive function⁶⁷, and early T2DM impairs hippocampal NVC in rats via reduced nitric oxide bioavailability⁶⁸. These findings reinforce altered haemodynamic signalling as a potential early driver of metabolically induced neurodegeneration.

Collectively, these findings support the hypothesis that neurovascular uncoupling and reduced CBF may serve as initiating events in obesity-induced neurodegeneration (Fig. 2). The brain may, at

least transiently, attempt to compensate via structural remodelling of vascular architecture, a response with both adaptive and pathological implications^{69,70}.

Structural reorganization of the cerebral vasculature

As the effector arm of NVC, the cerebral vasculature dynamically adjusts blood flow to match local metabolic demand⁷¹. This network, composed of endothelial cells, pericytes, smooth muscle cells and astrocyte endfeet, continuously interprets metabolic and inflammatory signals to maintain homeostasis. In obesity and neurodegenerative disease, however, this system undergoes profound remodelling, ranging from hypovascularization in vulnerable cortical regions to compensatory hypervascularization in metabolically active areas⁷².

Pathological angiogenesis is a well-recognized feature of neurodegeneration. In Alzheimer's disease, both amyloid- β and tau induce aberrant vascular growth: amyloid- β promotes angiogenesis and disrupts the BBB, whereas tau drives vessel narrowing, abnormal spiralling morphologies and increased but dysfunctional vessel density⁷³. These vessels often fail to deliver adequate nutrients and oxygen, creating microdomains of hypoxia that destabilize the NVU and amplify both neuronal and vascular injury. Such maladaptive angiogenesis reflects a breakdown in normal coupling between vascular remodelling and metabolic demand.

Obesity appears to engage similar pathways. In humans, BMI correlates positively with tortuosity of small cerebral arteries, a marker of pathological modelling associated with nearby white matter lesions⁷⁴. Post-mortem studies show increased arteriolar density in the hypothalamus of individuals with T2DM, consistent with region-specific angiogenesis⁷⁵. In murine models, a HFD or a high-fat high-sucrose (HFHS) diet induces increased vessel length and density, along with endothelial degeneration, particularly in the arcuate nucleus, the

master regulator of energy homeostasis⁷⁵. Hyperleptinaemia, a hallmark of obesity, may contribute to this effect through chronic activation of angiogenic signalling⁷⁶.

By contrast, the vascular architecture of higher-order cognitive regions remains poorly characterized, although reported reductions in CBF and NVC in these regions suggest progressive hypovascularization⁷⁷. Studies in rodents highlight dynamic, phase-dependent remodelling: early high-fat feeding increases vascular density and tortuosity in the frontal cortex⁷⁸, yet prolonged exposure produces vascular rarefaction in hippocampus and cortex⁷⁹. This pattern supports a model in which initial maladaptive hypervascularization gives way to progressive capillary loss, impairing nutrient delivery and accelerating cognitive decline.

Regional heterogeneity in remodelling suggests a broader organizing principle: brain areas at the brain–body interface (for example, hypothalamus and circumventricular organs) exhibit hypervascularization to enhance nutrient sensing, whereas higher-order cognitive regions, which lack such adaptive remodelling, become more susceptible to hypoperfusion and vulnerability. Irrespective of direction, sustained vascular dysfunction is closely linked to increased BBB permeability, immune activation and neuroinflammation⁸⁰.

BBB breakdown and immune cell infiltration

The BBB serves as both a protective filter and a communication interface between the brain and periphery⁸¹. BBB dysfunction is a hallmark of ageing and neurodegeneration, and growing evidence indicates that it has a central role in obesity-induced brain vulnerability⁸². In women with overweight or obesity, cerebrospinal fluid (CSF)-to-serum albumin ratio, an index of BBB permeability, is higher than in lean control women⁸³. Individuals with T2DM show similarly elevated BBB permeability to magnetic resonance imaging contrast agents⁸⁴, with serum albumin ratio positively correlating with CSF biomarkers of endothelial dysfunction⁸⁵. Post-mortem analysis reveals loss of tight-junction proteins and increased immune cell infiltration in T2DM brains, indicating widespread vascular leakage⁸⁶.

Interestingly, leptin transport across the BBB is impaired in obesity⁸⁷. Although this might look paradoxical at first, increased passive permeability to inert tracers and reduced active receptor-mediated transport of hormones are consistent with a compromised BBB. Chronic vascular stress thus creates a permissive environment for neurodegeneration.

The hippocampus is the region where BBB breakdown most consistently tracks with cognitive decline⁸⁸. Accumulating research suggests that the hippocampal BBB is more vulnerable to breakdown compared to other brain regions⁸⁹. Although studies examining hippocampal BBB integrity in humans with obesity are limited, preclinical studies indicate BBB dysfunction in obesity. In mice, 12–16 weeks of high-fat feeding result in increased tracer penetration and reduced tight-junction protein expression in hippocampus, whereas no abnormalities are detectable at 8 weeks⁹⁰. Remarkably, measurable increases in BBB permeability emerge within 1 h of high-fat exposure⁹¹, and 3 days of feeding induce permeability changes associated with learning and memory deficits⁹². These findings suggest a temporal sequence: (1) immediate functional permeability changes (hours–days), (2) compensatory stabilization period (weeks) and, (3) structural BBB breakdown (months).

A landmark study by Jais et al.⁹³ revealed a mechanistic basis for this progression. Short-term HFD feeding (3 days) downregulates GLUT1 expression in BBB endothelial cells, reducing glucose transport into the brain as a compensatory mechanism⁹³. Prolonged exposure (28 days) restores GLUT1 levels in conjunction with increased vascular endothelial growth factor (VEGF) expression in BBB-associated macrophages, suggesting an attempt to remodel the vasculature in response to nutrient stress. In chronic HFD conditions, myeloid-specific VEGF deletion further increases BBB permeability and decreases GLUT1 expression

in the cortex and hippocampus. These findings point to a model in which both hypervascularization and hypovascularization contribute to a leaky BBB, depending on regional vascular capacity and metabolic demand⁹⁴. Notably, hypothalamic regions, designed for dynamic nutrient sensing, may be more capable of compensatory vascular adaptation⁹⁵. By contrast, cortex and hippocampus, regions with more stable vascular structures, are less plastic and, therefore, more prone to dysfunction. This regional disparity may explain why certain brain areas are more vulnerable to obesity-induced neurodegeneration.

BBB breakdown permits the infiltration of peripheral immune cells, further altering the neuroimmune landscape⁹⁶. In mice, one week of a HFHS diet leads to infiltration of macrophages into the hypothalamus⁹⁷. Whereas changes in regulatory T cells occur more gradually, their decline over time correlates with increased type 1 helper T cells, shifting the immune balance towards chronic inflammation. Gain-of-function and loss-of-function studies confirm that regulatory T cells modulate hypothalamic inflammation and metabolic health⁹⁷. Similar infiltration occurs in cortex and septum⁹⁸, and correlates strongly with body weight and adipose inflammation.

These findings highlight a neuroinflammatory response to primary metabolic overload in the brain as an initial compensatory mechanism that becomes a chronic, secondary maladaptive response to metabolic dysfunction (Fig. 3). In summary, obesity-induced neuroinflammation is best understood as a two-phase process: an early, localized hypothalamic response to nutrient excess, followed by a secondary amplification driven by peripheral metabolic dysfunction and systemic inflammation.

This escalating immune system activation triggers downstream processes, including the release of matrix metalloproteases that degrade components of the ECM, setting the stage for structural disintegration and further functional decline⁹⁹.

ECM destabilization and PNN vulnerability

Once considered a passive structural scaffold, the ECM is now recognized as a dynamic regulator of the neural microenvironment, modulating synaptic plasticity, neuronal excitability and resistance to oxidative and mechanical stress¹⁰⁰. ECM dysfunction is increasingly implicated across neurological diseases, yet its role in metabolic overload is only beginning to be appreciated¹⁰¹. Emerging evidence suggests that endothelial cell remodelling may represent a critical mechanism linking obesity-induced inflammation to altered neural circuit function.

In the hypothalamus of obese mice, perineuronal nets (PNNs), specialized ECM structures surrounding neuronal soma, undergo augmentation and restructuring, a phenomenon recently described as neurofibrosis¹⁰². This remodelling is primarily driven by hypothalamic inflammation, which precedes and predicts metabolic dysfunction. Notably, PNN changes are concentrated around agouti-related peptide neurons in the arcuate nucleus, increasing their excitability and promoting hyperphagia¹⁰². These neurofibrotic changes not only enhance food intake but also impede insulin access to target neurons, contributing to central insulin resistance. Disruption of these aberrantly stabilized PNNs restores metabolic homeostasis and promotes weight loss, identifying ECM remodelling as a promising therapeutic target for metabolic diseases.

However, the relationship is not uniform. PNNs in the median eminence are required for maintaining energy balance in lean mice¹⁰³. This suggests that chronic metabolic overload may drive pathological amplification of PNNs, paralleling the maladaptive vascular remodelling seen in obesity. However, the role of obesity in regulation of hypothalamic PNNs remains unsettled as other studies in HFD-fed mice have found no increase in PNN formation in the arcuate nucleus¹⁰⁴. These discrepancies highlight methodological and temporal differences that require further clarification.

Beyond the hypothalamus, PNN-rich cortical and hippocampal circuits show distinct vulnerabilities. Short-term HFD or HFHS

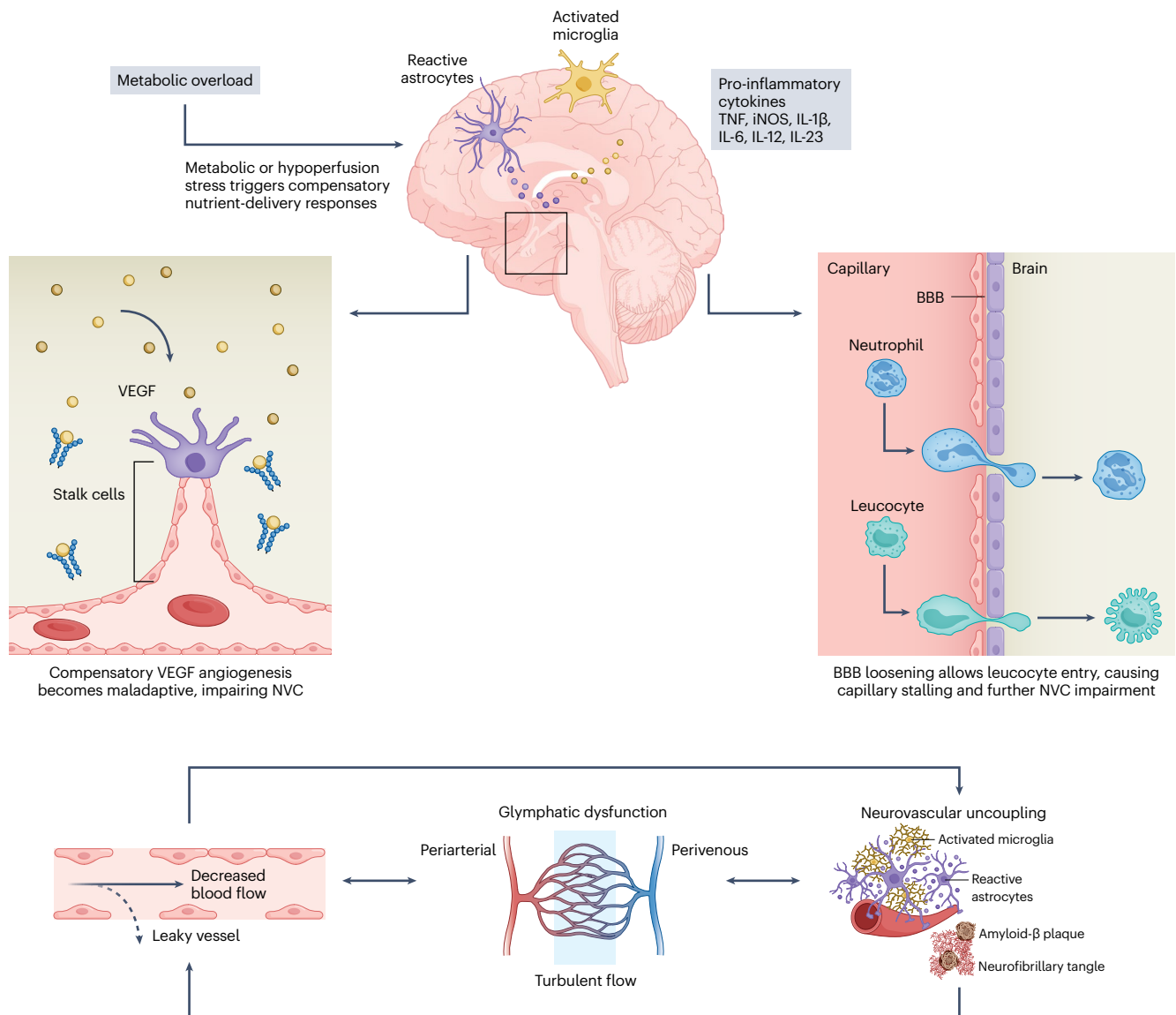


Fig. 3 | Network amplification and emergent compensation. Under metabolic or hypoperfusion stress, the brain mounts compensatory angiogenic and permeability responses to preserve nutrient delivery. In overnutrition, hypothalamic angiogenesis is driven by HIF1–VEGF signalling, which is consistent with a compensatory vascular response to metabolic stress. VEGF is a key signal that simultaneously increases BBB permeability and drives angiogenesis; however, VEGF signalling can also stall brain capillaries and impair microvascular flow. Newly formed vessels are often immature, exhibit reduced

mural cell coverage and show diminished vasoreactivity, which is a proxy for impaired NVC. These changes destabilize CBF regulation and disrupt AQP4 polarization. The combination of angiogenesis and BBB loosening facilitates leucocyte entry, which can further stall capillary flow and blunt NVC. Ensuing vascular inflammation further comprises glymphatic transport, creating a self-reinforcing loop of degeneration. IL, interleukin; iNOS, inducible nitric oxide synthase; TNF, tumour necrosis factor.

exposure reduces PNN intensity in cortex^{105,106} and decreases both parvalbumin interneuron numbers and PNN density in hippocampus¹⁰⁷. These effects appear age-dependent: adolescent mice exhibit relative resilience, whereas adults display rapid ECM degradation. Conversely, long-term HFD exposure (23 weeks) results in excessive hippocampal PNN accumulation, increased parvalbumin neuron number, impaired synaptic dysregulation and emotional dysregulation¹⁰⁸. Such opposing trajectories probably reflect stage-specific responses, initial ECM loss followed by reactive, maladaptive overproduction.

Taken together, these data suggest a bidirectional vulnerability: while hypothalamic circuits exhibit neurofibrotic PNN amplification that reinforces metabolic dysfunction, higher-order cognitive regions show PNN degradation or pathological accumulation, depending on exposure duration, contributing to cognitive impairment. Because the ECM regulates solute movement through the

interstitial space, its destabilization may also impair glymphatic clearance, promoting accumulation of neurotoxic metabolites such as amyloid- β and tau^{109,110}. This positions ECM remodelling as a central mediator in the bridge between metabolic overload and neurodegeneration.

Impaired glymphatic flow and accumulation of neurotoxic metabolites

The discovery of the glymphatic system reframed CSF flow as an active, regulated waste-clearance pathway essential for solute exchange and metabolic homeostasis¹¹¹. Glymphatic efficiency depends on sleep architecture, circadian rhythms, vascular pulsatility and astrocytic aquaporin 4 (AQP4) polarization, all of which are disrupted in obesity¹¹². These vulnerabilities make glymphatic function particularly sensitive to metabolic overload.

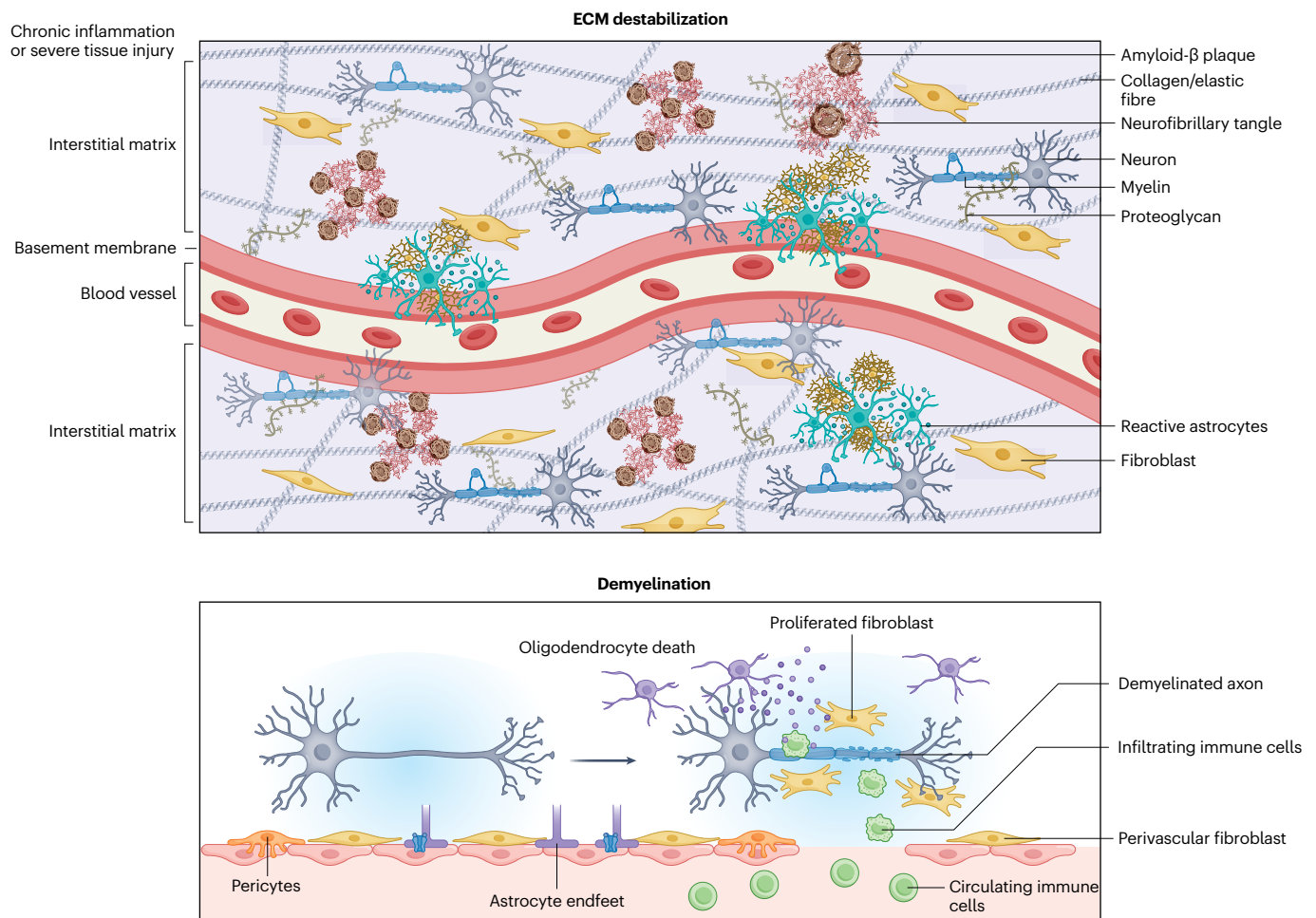


Fig. 4 | Cascade of structural and functional deterioration. Accumulation of neurotoxic molecules and pro-inflammatory immune cells can lead to both structural and functional breakdown. An inflammatory milieu induces perivascular fibroblasts and pericytes to expand and lay down a collagen-rich

scar that persists and disrupts tissue homeostasis. This results in a stiffer, leaky microvasculature that halts perivascular CSF exchange. The compounding effect of these processes creates a hostile microenvironment that jeopardizes oligodendrocyte survival and drives demyelination.

Human imaging studies link higher BMI to enlarged perivascular spaces and altered vasomotor oscillations across cerebellar, brainstem, insular and parietal regions¹¹³, patterns indicative of impaired interstitial waste clearance¹¹⁴. Glymphatic dysfunction is increasingly reported across metabolic and neurodegenerative conditions: in Parkinson's disease, higher BMI associates with reduced glymphatic function^{115,116}, and obesity-related idiopathic intracranial hypertension features increased extraventricular CSF volume and reduced venous outflow¹¹⁷. Parallel discoveries in meningeal lymphatics, including lymphangiogenesis near the cribriform plate and antigen sampling in dural lymphoid structures, underscore an active immunological interface that may be especially vulnerable to metabolic stress^{118,119}.

A seminal study by Delle et al.¹²⁰ demonstrated region-specific glymphatic remodelling in response to long-term HFD feeding. The hypothalamus showed increased glymphatic influx and enhanced AQP4 polarization, features that mirror its elevated vascularization and BBB permeability, and may represent an adaptive response to heightened metabolic demand. By contrast, glymphatic clearance and AQP4 polarization were reduced in cortex and hippocampus, aligning with their well-established vulnerability to cognitive decline¹²¹. Similar reductions in AQP4 levels across frontal–striatal circuits of obese mice, accompanied by gliosis and demyelination¹²², reinforce the interdependence of vascular, glial and fluid-transport pathways.

Animal models of T2DM further support these findings: hippocampal glymphatic flow is reduced in T2DM rats and correlates with

memory impairments¹²³. Although the relationship between specific metabolic phenotypes and glymphatic failure remains incompletely defined, the overarching principle is clear: altered metabolic states reshape brain-wide waste-clearance capacity¹¹⁶.

These findings align with a broader pattern: nutrient-sensing regions such as the hypothalamus can augment vascular and glymphatic function under nutrient excess. By contrast, areas with limited exposure to circulating signals (cognitive loci) lack this flexibility and may be predisposed to dysfunction under metabolic duress. Dysregulated glymphatic flow promotes accumulation of amyloid- β , tau and other neurotoxic metabolites, accelerating white matter damage and cognitive decline¹²⁴. Thus, glymphatic impairment represents a key mechanistic link between metabolic overload and progressive neurodegeneration.

Myelin breakdown and white matter structural decline

White matter loss has long been viewed as a downstream consequence of neurodegeneration, emerging only after substantial neuronal damage¹²⁵. However, growing evidence indicates that myelin disruption and axonal injury can arise early in disease progression, potentially initiating cognitive decline rather than merely reflecting it¹²⁶. Obesity increases the risk of neurological disorders well before clinical symptoms appear, with mounting evidence that alterations in brain myelin content contribute to this vulnerability¹²⁷.

BOX 1

Repurposing metabolic drugs for brain–body neurodegeneration

The purpose of viewing neurological disorders through a brain–body framework is to tackle the multifaceted nature of these complex pathologies. Owing to the interconnectedness between metabolic and neurological disorders, it is within reason to suggest that traditional metabolic drugs can be utilized for neurodegeneration. By preventing or treating metabolic dysfunctions with these metabolic drugs, they are indirectly mitigating the risk of neurological disorders. These drugs also demonstrate direct effects on the brain by targeting a variety of cell types and processes⁵³. In individuals with mild-to-moderate Alzheimer's disease, treatment with the GLP1R agonist liraglutide for 12 months reduces brain atrophy and cognition compared to the placebo group¹⁴². Similarly, the use of dipeptidyl peptidase 4 inhibitors reduces the rate of memory decline in people with Alzheimer's disease, whereas metformin improves memory performance in healthy control individuals¹⁴³. Treatment with GLP1R agonists or sodium-glucose cotransporter 2 inhibitors decreases the risk of Alzheimer's disease and related dementias in patients with T2DM¹⁴⁴.

In aged mice, metformin induces remyelination by restoring the regenerative capacity of aged oligodendrocytes¹⁴⁵. Linagliptin, a dipeptidyl peptidase 4 inhibitor, ameliorates cognitive impairment and brain insulin resistance in a rat model of Alzheimer's disease¹⁴⁶. A combination of canonical neurodegenerative therapies with complementary metabolic interventions may enhance treatment efficacy. For example, co-administration of metformin and donepezil improves cognitive function and abnormal glucose metabolism in individuals with non-dementia vascular cognitive impairment¹⁴⁷. The same combination improves brain antioxidant defences and attenuates hippocampal endoplasmic reticulum stress in diabetic rats¹⁴⁸. Although the data on human clinical trials are mixed, preclinical models demonstrate the potential of these metabolic drugs to improve neurological outcomes¹⁴⁹. Future optimization of the dose, timing and duration of treatment is required to maximize the therapeutic efficacy. It is also important to determine the optimal treatment window by identifying the proper patient populations that will most likely benefit from these treatments.

In cognitively unimpaired adults, increased BMI and central adiposity correlate with global reductions in myelin content, particularly within later-myelinating tracts such as the corona radiata, longitudinal fasciculus and thalamic radiations¹²⁸. These regions are highly susceptible to age-related degeneration, supporting the retrogenesis hypothesis, which posits that circuits myelinated late in development degenerate first under metabolic stress. Composite metabolic risk scores also predict lower myelin density in late-developing tracts¹²⁹. Visceral obesity is associated with increased white matter hyperintensities in the deep white matter, an association that persists independently of common obesity comorbidities such as hypertension¹³⁰. These microstructural changes correlate with reduced cognitive performance¹³¹.

Preclinical work begins to reveal underlying mechanisms. In mice, chronic HFD feeding induces hypothalamic demyelination, persistent microglial activation and elevated inflammatory cytokines¹³². These white matter abnormalities appear later than early immune and vascular changes, suggesting that demyelination may represent a downstream but integral phase of neurometabolic decline. Notably, reduced myelin content also occurs in the corpus callosum, frontal cortex and hippocampus, regions associated with cognition and interhemispheric communication¹³³.

Despite increased numbers of immature oligodendrocytes and upregulated myelin-related genes, these regenerative responses fail to restore structural integrity, suggesting a maladaptive or exhausted repair programme. Genetic models reinforce this principle: leptin-deficient *ob/ob* mice show profound hypomyelination and cognitive deficits¹³⁴, driven by chronic inflammation and the loss of leptin's pro-myelinating actions¹³⁵. Similar processes probably occur in diet-induced obesity, where leptin resistance compromises oligodendrocyte maturation and white matter stability.

In aged mice, visceral adiposity is associated with iron deposition and demyelination, supporting systemic inflammation as a key factor for driving neurodegeneration¹³⁶. Surgical removal of visceral fat reduces central iron levels and ameliorates myelin loss in the cortex and corpus callosum. Importantly, voluntary running, independent of weight loss, reverses many of the obesity-induced structural abnormalities in these regions, including demyelination and cerebrovascular dysfunction¹³³, underscoring the potential of behavioural interventions to restore white matter integrity even under persistent metabolic stress.

Taken together, these findings position white matter degradation as both a consequence and contributor to neurometabolic decline. In contrast to earlier, regionally heterogeneous changes, demyelination appears to propagate more uniformly across the brain, indicating that any sustained disruption to NVU function, whether through impaired glymphatic flow, vascular remodelling or immune infiltration, can compromise oligodendrocyte health and long-range network connectivity (Fig. 4).

Reframing neurodegeneration through a brain–body framework

Addressing the complexity of disorders such as obesity and neurodegeneration demands interdisciplinary perspectives. The emerging brain–body field reflects a transdisciplinary approach aimed at understanding the bidirectional interactions between central and peripheral systems¹³⁷. This framework has reshaped our understanding of metabolic and neurological disease by demonstrating that they share overlapping biological substrates and reciprocal regulatory loops, coining the 'neurometabolic' field.

A seminal example is the discovery of the hormone leptin, which catalysed a conceptual shift by revealing extensive brain–body cross-talk between adipose tissue and the brain¹³⁸. Leptin transformed obesity from a peripheral disorder of adiposity into a systemic condition involving long-range neural, endocrine and immunological circuits. Insights into the neurobiology of feeding behaviour and energy balance subsequently led to the development of glucagon-like peptide 1 receptor (GLP1R) agonists, which have revolutionized clinical management of obesity and T2DM¹³⁹. This same conceptual lens, identifying systemic metabolic drivers of neural dysfunction, is now increasingly applied to neurodegeneration. Preliminary data suggest that metabolic interventions such as GLP1R agonists and metformin may complement traditional neurological therapies^{140,141}, although whether their benefits arise from peripheral metabolic improvement, direct central nervous system actions or both remains an active area of investigation (Box 1).

The overarching message is integration. A neuron-centric view of neurodegeneration is no longer sufficient; instead, the full repertoire of central nervous system cell types (neurons, astrocytes, microglia, oligodendrocytes and endothelial cells) must be incorporated into models of disease. Similarly, rigid distinctions between 'central' and

‘peripheral’ disorders obscure the multifactorial nature of pathological processes and limit therapeutic discovery. Obesity, traditionally classified as a peripheral disorder, illustrates this principle: chronic metabolic overload reshapes neurovascular, immune and glial systems, thereby elevating long-term neurological risk.

A holistic brain–body framework opens avenues for intervention by acknowledging multiple interacting nodes of vulnerability (metabolic, vascular, inflammatory, synaptic and structural) that can be targeted to break self-reinforcing loops of degeneration.

Conclusion

Once regarded as separate clinical entities, obesity and neurological disorders are now understood to share deeply intertwined pathophysiological mechanisms. The obese brain exhibits features that map onto multiple models of neurodegenerative vulnerability: convergent initiation, cascade propagation, tipping-point transitions, network-level failure and, ultimately, multisystem homeostatic collapse. Chronic metabolic overload acts as a systemic stressor that perturbs neurons, glia, vasculature and immune cells. These disruptions are not isolated; instead, they interact dynamically across space and time, creating a self-reinforcing cycle of vulnerability. Importantly, metabolic stress increases the probability, not the inevitability, of neurodegenerative outcomes.

By integrating cascade and convergence models into a unified framework, we highlight where these processes intersect and how they evolve. This systems-level view reveals critical windows for intervention: blocking early triggers, slowing progressive cascades or interrupting feedback loops that amplify dysfunction.

Addressing the neurometabolic consequences of obesity will require strategies that extend beyond weight loss to target the cellular and molecular underpinnings of brain vulnerability. The fact that the system collapses only under multiple insults suggests that it may also be rescued through multiple therapeutic entry points. By elucidating the complex brain–body interactions that govern neurometabolic health, we can lay the foundation for more effective treatments for both metabolic and neurological diseases and ultimately bridge the gap between them.

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Author contributions

B.C. conceived the Perspective, developed the conceptual framework, performed the literature synthesis and wrote the original manuscript. M.S. and E.T. provided critical feedback and supervision. A.R.-D. assisted with figure design and preparation. All authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence should be addressed to Bandy Chen, Marc Schneeberger or Eric Topol.

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